

a tripod and using the stopwatch to measure time. I did observations to determine how repeatable the ball motion was. If it wasn't repeatable, it would mean there was too much randomness for the process to work. I plotted charts of where the ball was at various points in time. The results looked pretty good; the process did seem repeatable.

I also created an analog experiment by letting a roulette ball slide down an inclined track across the floor. The idea was that I could translate the rotary motion of the wheel into a linear equivalent. I could take the potential energy at the height at which I launched the ball as the amount of kinetic energy the ball was going to get when it rolled down. I wanted to see if the same amount of potential energy converted to kinetic energy would send the ball to approximately the same place on the floor each time. It did. It wasn't proof, but it told me that I couldn't rule out the possibility that predicting roulette might work. One night, my wife had invited my in-laws over to dinner, and I was so engrossed in my experiments that I lost track of what was going on around me. They wondered where I was because I didn't show up at the dinner table. They came looking for me and found me rolling balls down the track. I'm sure that, at that moment, they must have thought that their daughter had made a very serious mistake.

Or that you had lost your marbles. How does this tie back to Shannon?

When Shannon asked what else I was working on, I started telling him about my roulette experiments. Shannon was the ultimate gadgeteer. He got very excited because this idea was right up his alley. The few minutes I had been allocated stretched into several hours. We decided to continue the project together. We ordered a reconditioned wheel from Vegas. I remember it cost \$1,500. There was great excitement when it arrived at Shannon's house in a huge packing case. We set it up in Shannon's basement on a slate pool table, which was rock solid. We got strobe equipment from MIT so that when you spun the ball, you could flash the strobe, which would light up briefly, making the ball look stationary. The effect was like a strobe light in a disco. We also had a large clock with a second hand that made one revolution per second, with the dial divided into hundredths of a second. We were able to hit the strobe and stop the